

Project Cost & Risk Management Analysis



SMART EXPENSE SPLITTER

Project	Smart Expense Splitter with Advanced Receipt Scanner
Project Sponsor	Course Instructor
Project Manager	Mayisha Hossain Bhuiyan (2026-2-96-020)
Course	CSE596: IT Project Management & Entrepreneurship
Project Duration	01 July 2026 - 31 December 2026 (6 Months)
Assumed Status Date	31 October 2026
Initial Direct Cost	\$100,000
Total Approved Budget	\$115,000

Version	Date	Author	Approved By
V1.2	31 October 2026	PM	Instructor

Project Overview

The Smart Expense Splitter with Advanced Receipt Scanner is a mobile and web-based application designed to simplify the process of dividing shared expenses. The system allows users to scan or upload receipts, extract item names and prices using OCR, assign individual or shared items to participants, and automatically calculate how much each person should pay. The project aims to reduce manual calculation errors, save time, and improve transparency when splitting group expenses.

Project Objective

The main objectives of the project are to:

- Develop a user-friendly mobile and web application for managing shared expenses.
- Allow users to create groups and manage participants.
- Scan and upload receipts using OCR technology.
- Allow users to manually correct OCR results when necessary.
- Assign receipt items to one person, selected participants, or the whole group.
- Automatically calculate each participant's share.
- Generate a clear payment summary showing who paid, who owes, and how much needs to be settled.
- Maintain expense history and support reminders or notifications.

Gantt Chart

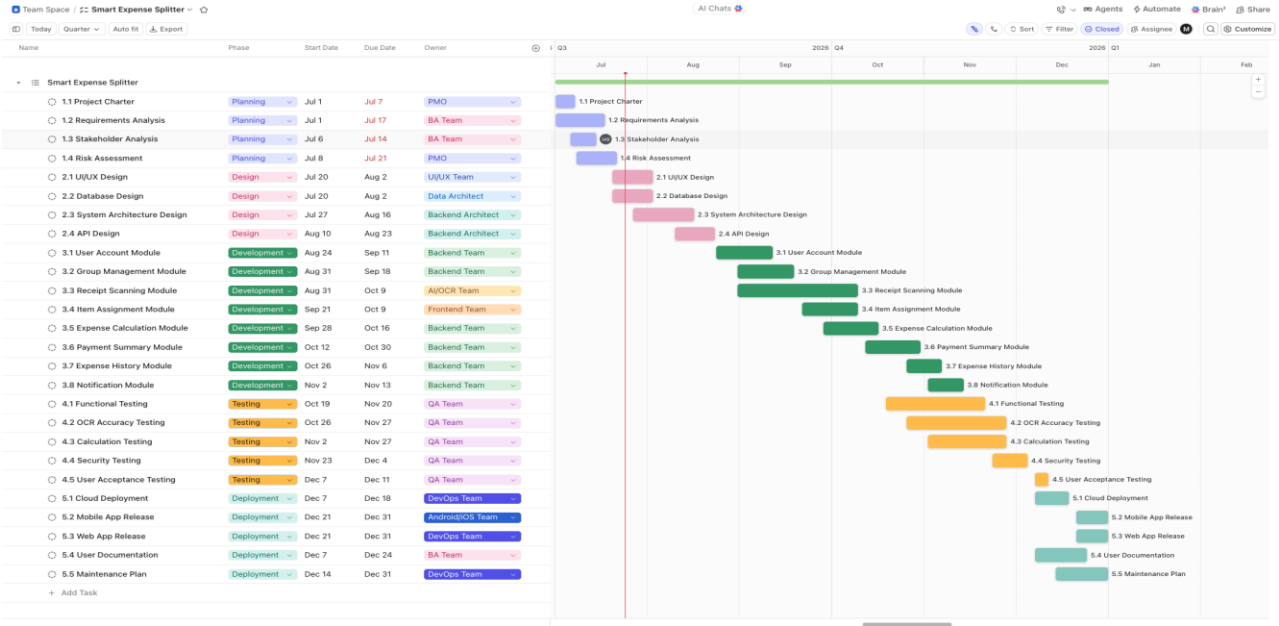


Figure 1. Project Gantt Chart for the Smart Expense Splitter

The Gantt chart in the Figure 1 presents the six-month project schedule from July to December 2026, showing the timing and overlap of the major phases: Planning, Design, Development, Testing, and Deployment. It also shows the sequence, duration, and assigned team for individual project tasks, helping track project progress and dependencies.

Project Status Assumption

The Smart Expense Splitter project is assumed to be in execution as of 31 October 2026. Planning and design are complete, development is still active, and early testing has started. The project has experienced some delay and extra spending mainly because of OCR integration, receipt-scanning accuracy issues, and additional debugging.

Part A: Project Cost Management

1. Cost Breakdown

A bottom-up cost estimate is used by assigning a budget to each project task. The total direct cost is \$100,000.

WBS	Task	Budget	Percentage (%)
1.1	Project Charter	\$2,000	2%
1.2	Requirements Analysis	\$3,000	3%
1.3	Stakeholder Analysis	\$1,500	1.5%
1.4	Risk Assessment	\$1,500	1.5%
	Planning Total	\$8,000	8%
2.1	UI/UX Design	\$4,000	4%
2.2	Database Design	\$3,000	3%
2.3	System Architecture Design	\$3,000	3%
2.4	API Design	\$2,000	2%
	Design Total	\$12,000	12%
3.1	User Account Module	\$5,000	5%
3.2	Group Management Module	\$5,000	5%
3.3	Receipt Scanning/OCR Module	\$12,000	12%
3.4	Item Assignment Module	\$5,000	5%
3.5	Expense Calculation Module	\$7,000	7%
3.6	Payment Summary Module	\$5,000	5%
3.7	Expense History Module	\$4,000	4%
3.8	Notification Module	\$5,000	5%
	Development Total	\$48,000	48%
4.1	Functional Testing	\$4,000	4%
4.2	OCR Accuracy Testing	\$4,000	4%
4.3	Calculation Testing	\$3,500	3.5%
4.4	Security Testing	\$3,500	3.5%
4.5	User Acceptance Testing	\$2,000	2%
	Testing Total	\$17,000	17%
5.1	Cloud Deployment	\$5,000	5%
5.2	Mobile App Release	\$3,000	3%
5.3	Web App Release	\$2,500	2.5%
5.4	User Documentation	\$2,000	2%
5.5	Maintenance Plan	\$2,500	2.5%
	Deployment Total	\$15,000	15%
	TOTAL DIRECT COST	\$100,000	100%

2. Contingency & Management Reserve

Following the approach demonstrated in the course material, a 5% contingency reserve is kept for identified risks and a 10% management reserve is kept for unknown or unexpected risks.

Contingency Reserve: $\$100,000 \times 5\% = \$5,000$

Management Reserve: $\$100,000 \times 10\% = \$10,000$

Budget Component	Amount
Direct Cost	\$100,000
Contingency Reserve	\$5,000
Cost Baseline	\$105,000
Management Reserve	\$10,000
Total Approved Project Budget	\$115,000

For Earned Value Management (EVM), the cost baseline is \$105,000. The \$10,000 management reserve is not included in normal performance measurement.

3. Earned Value Management (EVM) - Status at 31 October 2026

Planned Value (PV)

Planned Value represents the budgeted value of the work that should have been completed by the status date.

PV: Budgeted value of scheduled work = \$63,400

Earned Value (EV)

Earned Value represents the budgeted value of the work actually completed by 31 October 2026.

Completed Work	Earned Value
Planning	\$8,000
Design	\$12,000
User Account Module	\$5,000
Group Management Module	\$5,000
Receipt Scanner - 80%	\$9,600
Item Assignment - 90%	\$4,500
Expense Calculation - 85%	\$5,950
Payment Summary - 70%	\$3,500
Expense History - 20%	\$800
Functional Testing	\$1,000
OCR Testing	\$400
TOTAL EV	\$55,750

EV: Budgeted value of completed work = \$55,750

Actual Cost (AC)

Actual Cost is the real amount spent for the work performed. Extra OCR API usage and debugging increased spending.

AC: Actual expenditure to date = \$60,000

The graph in Figure 2 compares Planned Value (PV), Earned Value (EV), and Actual Cost (AC) throughout the project timeline. At the October status date, PV = \$63,400, EV = \$55,750, and AC = \$60,000. Since EV is lower than PV, the project is behind schedule, and since AC is higher than EV, the project is over budget.

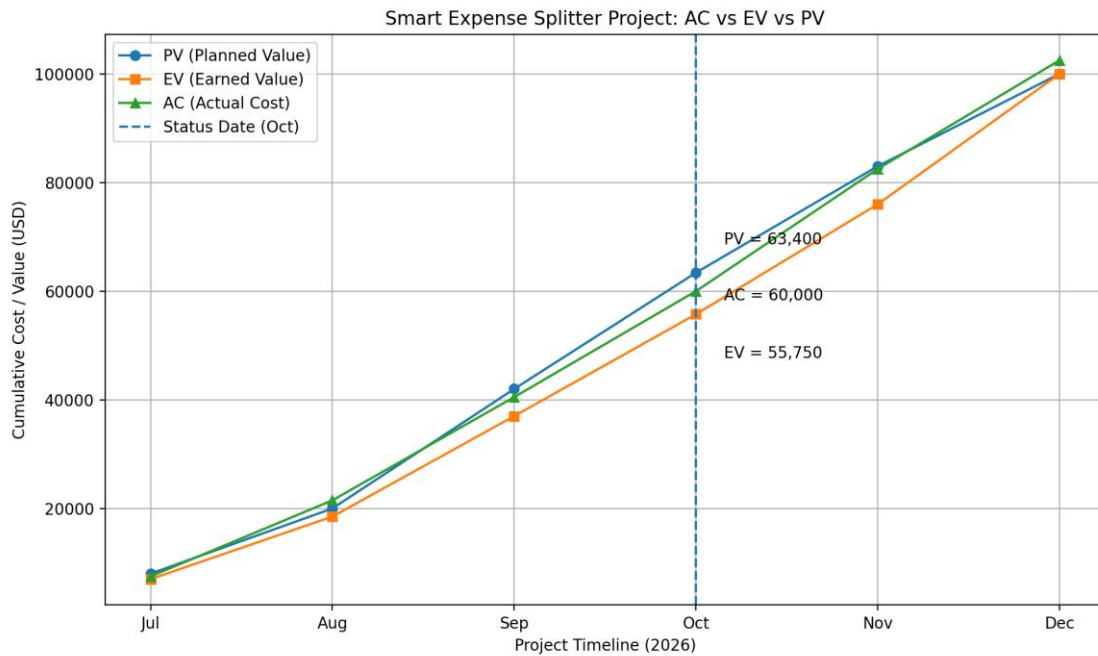


Figure 2. Earned Value Management (EVM) Performance of the Smart Expense Splitter Project

4. EVM Calculations

Cost Variance (CV): $EV - AC = \$55,750 - \$60,000 = -\$4,250$

Interpretation: CV is negative, so the project is over budget by \$4,250 for the amount of work completed.

Schedule Variance (SV): $EV - PV = \$55,750 - \$63,400 = -\$7,650$

Interpretation: SV is negative, so the project is behind schedule.

Cost Performance Index (CPI): $EV / AC = \$55,750 / \$60,000 = 0.93$

Interpretation: $CPI < 1$ means cost performance is inefficient. For every \$1.00 spent, about \$0.93 of planned value is earned.

Schedule Performance Index (SPI): $EV / PV = \$55,750 / \$63,400 = 0.88$

Interpretation: $SPI < 1$ means the project is behind schedule and is progressing at about 88% of the planned rate.

5. EVM Summary

Metric	Value	Meaning
PV	\$63,400	Work planned by the status date
EV	\$55,750	Budgeted value of work actually completed
AC	\$60,000	Actual money spent
CV	-\$4,250	Over budget
SV	-\$7,650	Behind schedule
CPI	0.93	Over budget
SPI	0.88	Behind schedule

6. Corrective Actions

- Reduce unnecessary OCR API calls and set usage limits.
- Compare alternative OCR and cloud service providers.
- Freeze non-essential feature additions until schedule performance improves.
- Reassign available developers from completed modules to delayed modules.
- Give priority to the Receipt Scanning and Expense Calculation modules.
- Run development and testing in parallel where practical.
- Review cost and schedule performance every week using EVM.

Part B: Project Risk Management

1. Risk Identification

The key project risks were identified from the original project charter and organized for the current execution stage.

ID	Risk
R1	OCR may incorrectly extract receipt items or prices.
R2	Scope changes may delay development.
R3	Expense calculation defects may produce incorrect splits.
R4	OCR or cloud service costs may increase.
R5	Security or privacy weaknesses may expose user data.
R6	App or cloud deployment may be delayed.

2. Risk Breakdown Structure (RBS)

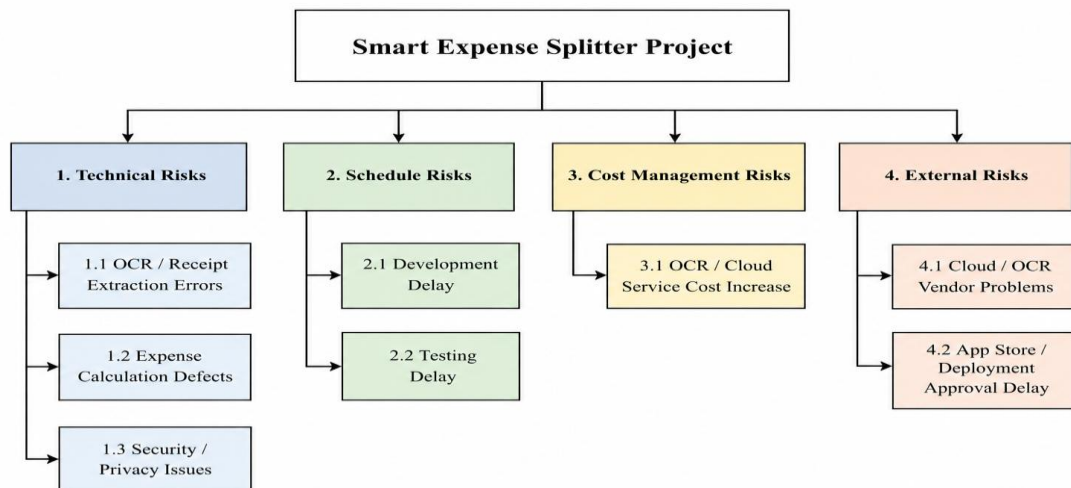


Figure 3. Risk Breakdown Structure of the Smart Expense Splitter Project

The Figure 3 organizes the project risks into four main categories: Technical, Schedule, Cost Management, and External Risks. Each category is further divided into specific risks such as OCR extraction errors, calculation defects, development delays, cloud cost increases, vendor problems, and deployment approval delays.

3. Risk Probability, Impact & Score

Here, a 1-5 scale is used.

Risk Score = Probability Rating × Impact Rating.

Rating	Probability	Impact
1	Very Low	Very Low
2	Low	Low
3	Medium	Medium
4	High	High
5	Very High	Very High

4. Risk Register

ID	Risk	Probability	Impact	Score	Level	Owner
R1	OCR errors	4	5	20	High	AI/OCR Lead
R2	Scope changes	3	4	12	Medium-High	Project Manager
R3	Calculation defects	3	5	15	High	Backend / QA
R4	Cloud/OCR cost increase	3	3	9	Medium	PM / DevOps
R5	Security/privacy issue	2	5	10	Medium-High	Security / QA
R6	Deployment delay	2	3	6	Medium	DevOps

The highest-priority risks are R1 (OCR errors, score 20) and R3 (calculation defects, score 15).

5. Impact Analysis

Risk	Cost Impact	Schedule Impact	Quality Impact
OCR Error	High	High	High
Scope Creep	High	High	Medium
Calculation Error	Medium	Medium	Very High
Cloud Cost Increase	High	Low	Low
Security Issue	High	Medium	Very High
Deployment Delay	Medium	High	Low

6. Quantitative Risk Analysis (EMV)

Expected Monetary Value (EMV) = Probability × Financial Impact

Risk	Probability	Financial Impact
OCR errors	40%	\$5,000
Scope change	30%	\$4,000
Calculation defects	20%	\$3,000
Cloud/OCR cost increase	25%	\$2,400
Security issue	10%	\$4,000
Deployment delay	20%	\$1,000

$$\text{EMV}_1: 0.40 \times \$5,000 = \$2,000$$

$$\text{EMV}_2: 0.30 \times \$4,000 = \$1,200$$

$$\text{EMV}_3: 0.20 \times \$3,000 = \$600$$

$$\text{EMV}_4: 0.25 \times \$2,400 = \$600$$

$$\text{EMV}_5: 0.10 \times \$4,000 = \$400$$

$$\text{EMV}_6: 0.20 \times \$1,000 = \$200$$

$$\text{Total EMV: } \$2,000 + \$1,200 + \$600 + \$600 + \$400 + \$200 = \$5,000$$

Interpretation: The project's total expected financial risk exposure is \$5,000. This matches the \$5,000 contingency reserve, so the proposed contingency reserve is appropriate for the identified risks.

7. Risk Response Plan

Risk	Strategy	Response Action
OCR errors	Mitigate	Test multiple receipt types, compare OCR vendors, and allow manual correction.
Scope changes	Avoid	Freeze the approved MVP scope. Any non-essential new feature request will be postponed to a future version instead of adding it to the current project.
Calculation defects	Mitigate	Use unit tests, edge-case tests, and independent formula review.
Cloud/OCR cost increase	Mitigate / Transfer	Use quotas, compare vendors, and prefer predictable pricing.
Security issue	Mitigate	Use authentication, encryption, least privilege, and security testing.
Deployment delay	Mitigate	Start app-store and cloud deployment preparation early, use a deployment checklist, and verify all release requirements before the scheduled release date.

8. Contingency Planning

A risk response tries to reduce the probability or impact before the event happens. A contingency plan defines what the team will do if the risk actually occurs.

Risk	Trigger	Contingency Plan
OCR failure	OCR accuracy drops below 90%	Switch to backup OCR service and require manual correction.
Scope creep	A major new feature is requested	Defer the feature to the next release.
Calculation defects	Critical calculation test fails	Stop release and repair the calculation module.
Cost overrun	OCR/cloud usage exceeds the defined threshold	Use contingency reserve and change service plan.
Security problem	A serious vulnerability is found	Disable affected feature, patch, retest, and review access.
Deployment delay	Release or approval fails	Release the web version first while mobile approval continues.

9. Project Risk Monitoring

Risk management should continue throughout the project rather than happening only once.

Monitoring Activity	Frequency
Review risk register	Weekly
Review high-priority risks	Every project meeting
Check OCR accuracy	Every testing cycle
Check OCR/cloud cost usage	Weekly
Identify new risks	Weekly
Security review	Before release

Deployment readiness review	Before final deployment
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The Project Manager should update probability, impact, risk owner, response status, newly identified risks, and closed risks throughout the project.

10. Risk Recommendations

- Prioritize OCR and calculation risks because they directly affect the core product.
- Maintain the \$5,000 contingency reserve for identified risks.
- Keep the \$10,000 management reserve for unknown or unexpected risks.
- Freeze non-essential functionality until the schedule improves.
- Conduct weekly EVM and risk reviews.
- Keep manual OCR correction as a fallback.
- Complete security testing before deployment.
- Start app/cloud release preparation early.

Part C: Final Recommendation

Based on the cost and risk analysis, the Smart Expense Splitter project should continue, but corrective actions are required. At the current status date, the project has a CPI of 0.93 and an SPI of 0.88, indicating that it is over budget and behind schedule. The most significant risks are OCR extraction errors and expense calculation defects. The project's total EMV is \$5,000, which is covered by the proposed \$5,000 contingency reserve, while an additional \$10,000 management reserve is maintained for unexpected risks. Therefore, the project remains feasible, but the Project Manager should control scope, prioritize delayed core modules, closely monitor OCR and calculation quality, and conduct weekly cost and risk reviews to complete the project within the approved budget and schedule.

Key Financial and Performance Indicators

Item	Value
Direct Cost	\$100,000
Contingency Reserve	\$5,000
Management Reserve	\$10,000
Cost Baseline	\$105,000
Total Approved Budget	\$115,000
PV	\$63,400
EV	\$55,750
AC	\$60,000
CV	-\$4,250
SV	-\$7,650
CPI	0.93
SPI	0.88
Total EMV	\$5,000